

Q1.

Alex and Sam are playing a zero-sum game.

The game is represented by the pay-off matrix for Alex.

		Sam		
Alex	Strategy	S₁	S₂	S₃
	A₁	2	2	3
	A₂	0	3	5
	A₃	-1	2	-2

- (a) Explain why the value of the game is 2

(3)

- (b) Identify the play-safe strategy for each player.

(1)

(Total 4 marks)

Q2.

Harry and Will play a zero-sum game. The game is represented by the following pay-off matrix for Harry.

		Will			
Harry	<i>Strategy</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>
	<i>A</i>	4	-1	2	3
	<i>B</i>	4	6	3	7
	<i>C</i>	1	3	-2	4

- (a) Show that this game has a stable solution and state the play-safe strategy for each player.

(4)

- (b) List any saddle points.

(1)

(Total 5 marks)

Q3.

- (a) Two people, Adam and Bill, play a zero-sum game. The game is represented by the following pay-off matrix for Adam.

		Bill		
Adam	<i>Strategy</i>	B₁	B₂	B₃
	A₁	−6	−1	−5
	A₂	5	2	−3
	A₃	−5	4	−4



A_4	2	1	-4
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- (i) Show that this game has a stable solution. (3)
- (ii) Find the play-safe strategy for each player. (1)
- (iii) State the value of the game for **Bill**. (1)
- (b) Roza plays a different zero-sum game against a computer. The game is represented by the following pay-off matrix for Roza.

		Computer			
		Strategy	C ₁	C ₂	C ₃
Roza	R ₁		3	4	-3
	R ₂		-2	-1	5

- (i) State which strategy the computer should never play, giving a reason for your answer. (1)
- (ii) Roza chooses strategy R_1 with probability p . Find expressions for the expected gains for Roza when the computer chooses each of its two remaining strategies. (2)
- (iii) Hence find the value of p for which Roza will maximise her expected gains. (2)
- (iv) Find the value of the game for Roza. (1)

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(Total 11 marks)

Q4.

- (a) Two people, Ann and Bill, play a zero-sum game. The game is represented by the following pay-off matrix for Ann.

		Bill		
Ann	Strategy	B ₁	B ₂	B ₃
	A ₁	−1	0	−2
	A ₂	4	−2	−3
	A ₃	−4	−5	−3

Show that this game has a stable solution and state the play-safe strategies for Ann and Bill.

(4)

- (b) Russ and Carlos play a different zero-sum game, which does not have a stable solution. The game is represented by the following pay-off matrix for Russ.

		Carlos			
		<i>Strategy</i>	C ₁	C ₂	C ₃
Russ	R ₁	−4	7	−3	
	R ₂	2	−1	1	

- (i) Find the optimal mixed strategy for Russ.

(7)

- (ii) Find the value of the game.

(1)

(Total 12 marks)

Q5.

- (a) Two people, Raj and Cal, play a zero-sum game. The game is represented by the following pay-off matrix for Raj.

		Cal			
		Strategy	X	Y	Z
Raj	I	-7	8	-5	
	II	6	2	-1	
	III	-2	4	-3	

Show that this game has a stable solution and state the play-safe strategy for each player.

(4)

- (b) Ros and Carly play a different zero-sum game for which there is no stable solution. The game is represented by the following pay-off matrix for Ros, where x is a constant.

		Carly	
		<i>Strategy</i>	
		C₁	C₂
ROs	R₁	5	x
	R₂	-2	4

Ros chooses strategy R_1 with probability p .

- (i) Find expressions for the expected gains for Ros when Carly chooses each of the strategies C_1 and C_2 .

(2)

- (ii) Given that the value of the game is $\frac{8}{3}$, find the value of p and the value of x .

(4)

(Total 10 marks)

Q6.

Two people, Rob and Con, play a zero-sum game.

The game is represented by the following pay-off matrix for Rob.

		Con			
		Strategy	C ₁	C ₂	C ₃
Rob	R ₁		−2	5	3
	R ₂		3	−3	−1
	R ₃		−3	3	2

- (a) Explain what is meant by the term 'zero-sum game'.

(1)

- (b) Show that this game has no stable solution.

(3)

- (c) Explain why Rob should never play strategy R_3 .

(1)

- (d) (i) Find the optimal mixed strategy for Rob.

(7)

- (ii) Find the value of the game.

(1)

(Total 13 marks)

Q7.

- (a) Two people, Ros and Col, play a zero-sum game. The game is represented by the following pay-off matrix for Ros.

		Col		
	Strategy	X	Y	Z
	I	−4	−3	0
	II	5	−2	2

Ros	III	1	-1	3
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- (i) Show that this game has a stable solution. (3)
- (ii) Find the play-safe strategy for each player and state the value of the game. (2)
- (b) Ros and Col play a different zero-sum game for which there is no stable solution. The game is represented by the following pay-off matrix for Ros.

		Col		
		C ₁	C ₂	C ₃
Ros	R ₁	3	2	1
	R ₂	-2	-1	2

- (i) Find the optimal mixed strategy for Ros. (7)
- (ii) Calculate the value of the game. (1)
- (Total 13 marks)

Q8.

- (a) Afzal and Bill play a zero-sum game. The game is represented by the following pay-off matrix for Afzal.

		Bill			
		I	II	III	IV
Afzal	I	3	-2	1	2
	II	3	5	2	6
	III	0	2	-3	3

- (i) Show that this game has a stable solution. (3)
- (ii) List any saddle points. (1)
- (b) Colin and David play a zero-sum game. The game is represented by the following pay-off matrix for Colin.

		David	
		I	II

Colin

I	$x + 2$	$x - 1$
II	3	5

Given that the optimal mixed strategy makes the value of the game $\frac{19}{5}$,
find the value of x .

(8)

(Total 12 marks)

Q9.

Alma and Brian play a zero-sum game. The game is represented by the following pay-off matrix for Alma.

		Brian			
		I	II	III	IV
Alma	I	3	-1	7	-7
	II	3	-6	0	8
	III	8	12	8	11
	IV	8	13	8	10

(a) Write down the pay-off for **Brian** if Alma plays I and Brian plays IV.

(1)

(b) Show that this game has a stable solution.

(3)

(c) Find all the saddle points for the game.

(2)

(Total 6 marks)

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